COMPLETE TRADING ENCYCLOPEDIA

Financial Instruments, Technical Indicators, Patterns, Methods and Metrics

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Note: Yeh educational reference hai. Koi bhi instrument ya indicator profit guarantee nahi karta. Naye instruments har saal aate hain, isliye 'har cheez' ki final list possible nahi, lekin yeh ab tak ki sab se badi practical collection hai. Defaults (jaise RSI 14) common settings hain, aap change kar sakte hain.

PART 1: FINANCIAL INSTRUMENTS

1.1 World Currencies Used in Forex (ISO codes)

Group	Currencies
G10	USD (US Dollar), EUR (Euro), JPY (Yen), GBP (Pound), CHF (Swiss Franc), AUD, NZD, CAD, SEK (Swedish Krona), NOK (Norwegian Krone)
Asia	CNY/CNH (Yuan), HKD, SGD, INR, PKR, BDT, LKR, NPR, IDR, MYR, THB, PHP, VND, KRW, TWD
Middle East / Africa	AED, SAR, QAR, KWD, BHD, OMR, ILS, TRY, EGP, ZAR, NGN, KES, GHS, MAD
Europe (non-euro)	PLN, CZK, HUF, RON, DKK, ISK, RUB, UAH, RSD
Americas	MXN, BRL, ARS, CLP, COP, PEN

1.2 Forex Major Pairs (with USD, most liquid)

Pair	Nickname	Characteristics
EUR/USD	Fiber	Most traded pair in the world, tight spread, active London/NY
USD/JPY	Gopher / Ninja	Sensitive to US yields and BoJ policy, safe haven flows
GBP/USD	Cable	Volatile, UK data and BoE driven
USD/CHF	Swissie	Safe haven, inverse correlation with EUR/USD
AUD/USD	Aussie	Commodity and China risk proxy
USD/CAD	Loonie	Oil correlated
NZD/USD	Kiwi	Dairy and risk sentiment

1.3 Forex Minor / Cross Pairs

Base	Pairs
EUR	EUR/GBP, EUR/JPY, EUR/CHF, EUR/AUD, EUR/CAD, EUR/NZD
GBP	GBP/JPY (Guppy/Dragon), GBP/CHF, GBP/AUD, GBP/CAD, GBP/NZD
AUD	AUD/JPY, AUD/CHF, AUD/CAD, AUD/NZD
NZD	NZD/JPY, NZD/CHF, NZD/CAD
CAD / CHF	CAD/JPY, CAD/CHF, CHF/JPY

1.4 Forex Exotic Pairs

Region	Pairs
USD vs Asia	USD/CNH, USD/CNY, USD/HKD, USD/SGD, USD/INR, USD/PKR, USD/BDT, USD/LKR, USD/IDR, USD/MYR, USD/THB, USD/PHP, USD/KRW, USD/TWD, USD/VND
USD vs Middle East / Africa	USD/TRY, USD/ILS, USD/AED, USD/SAR, USD/QAR, USD/KWD, USD/EGP, USD/ZAR, USD/NGN, USD/KES, USD/MAD
USD vs Europe	USD/PLN, USD/CZK, USD/HUF, USD/RON, USD/SEK, USD/NOK, USD/DKK, USD/RUB, USD/UAH
USD vs Americas	USD/MXN, USD/BRL, USD/ARS, USD/CLP, USD/COP, USD/PEN
EUR exotics	EUR/TRY, EUR/ZAR, EUR/PLN, EUR/CZK, EUR/HUF, EUR/RON, EUR/SEK, EUR/NOK, EUR/DKK, EUR/MXN, EUR/ILS, EUR/SGD, EUR/HKD
GBP exotics	GBP/ZAR, GBP/TRY, GBP/PLN, GBP/SEK, GBP/NOK, GBP/DKK, GBP/MXN, GBP/SGD
Other	SGD/JPY, ZAR/JPY, NOK/SEK, TRY/JPY, MXN/JPY, CNH/JPY, AUD/SGD, CHF/SGD

1.5 Other FX Instruments

Instrument	Description
Spot FX	Immediate exchange (T+2, T+1 for USD/CAD)
FX Forwards / Outright Forwards	Fixed future rate
FX Swaps / Currency Swaps	Exchange and reverse exchange at set dates
NDF (Non-Deliverable Forwards)	Cash settled forwards for restricted currencies (INR, CNY, KRW, BRL, TWD, IDR)
FX Futures (CME 6E, 6J, 6B, 6A, 6C, 6S, 6N, 6M)	Exchange traded currency futures

Instrument	Description
FX Options	Vanilla and exotic (barrier, digital, one-touch, range, Asian, lookback)
Currency Indices	DXY (USD Index), EXY (Euro Index), BXY (GBP), JXY (JPY), AXY (AUD), CXY (CAD), SXY (CHF), ZXY (NZD)
Currency Strength Meters	Custom baskets (8 major currencies)
Crypto/Fiat Pairs	BTC/USD, ETH/EUR, USDT/PKR etc.
Metal/Fiat Pairs	XAU/USD, XAU/EUR, XAU/GBP, XAU/AUD, XAU/CHF, XAU/JPY, XAG/USD, XAG/EUR, XAU/XAG (ratio)

1.6 Precious and Industrial Metals

Symbol	Instrument	Notes
XAU	Gold	Safe haven, inflation hedge, quoted per troy oz
XAG	Silver	Industrial plus monetary, more volatile than gold
XPT	Platinum	Auto catalysts, jewelry
XPD	Palladium	Auto catalysts (gasoline)
XRH / Rhodium, Iridium, Ruthenium, Osmium	Rare platinum group metals	OTC, very illiquid
HG (COMEX)	Copper	Economic health barometer ('Dr. Copper')
LME Aluminium	Aluminium	Packaging, auto, aerospace
LME Zinc, Nickel, Lead, Tin	Base metals	Galvanizing, batteries, solder
Cobalt, Lithium (carbonate/hydroxide), Manganese	Battery metals	EV supply chain
Iron Ore (SGX/DCE), Steel (HRC, Rebar)	Steel chain	China driven
Rare Earths (Neodymium, Dysprosium, etc.)	Specialty	Magnets, electronics
Uranium (U3O8)	Nuclear fuel	Spot and term pricing
Titanium, Molybdenum, Vanadium, Tungsten, Gallium, Germanium	Minor metals	Strategic materials

1.7 Energy

Symbol	Instrument	Notes
WTI (CL)	US Light Sweet Crude	NYMEX benchmark, Cushing delivery
Brent (BZ / LCO)	North Sea Crude	Global benchmark (ICE)
Dubai / Oman / Murban	Middle East benchmarks	Asia pricing
OPEC Basket, Urals, WCS, Tapis	Other crude grades	
NG (Henry Hub)	US Natural Gas	Weather and storage driven
TTF, NBP, JKM	European, UK, Asian gas	LNG benchmarks
RB (RBOB Gasoline), HO (Heating Oil), ULSD, Gasoil, Jet Fuel	Refined products	Crack spreads
Propane, Ethane, Butane	NGLs	
Coal (Newcastle, Rotterdam, API2)	Thermal coal	
Electricity (PJM, Nord Pool, EEX)	Power futures	Regional
EUA, UKA, CCA, RGGI	Carbon emission allowances	Compliance carbon markets
Renewable Energy Certificates (RECs)	Green certificates	

1.8 Agricultural Commodities

Group	Instruments
Grains	Corn, Wheat (Chicago SRW, KC HRW, MGEX Spring), Soybeans, Soybean Meal, Soybean Oil, Oats, Rough Rice, Barley, Rapeseed/Canola, Sorghum
Softs	Coffee (Arabica, Robusta), Cocoa, Sugar #11, White Sugar, Cotton #2, Orange Juice (FCOJ)
Livestock and Dairy	Live Cattle, Feeder Cattle, Lean Hogs, Class III Milk, Butter, Cheese, Whey, Pork Bellies (historical)
Fibers / Others	Lumber, Rubber, Wool, Palm Oil, Coconut Oil, Tea, Potatoes, Eggs, Fertilizer (Urea, DAP, Potash), Ethanol, Oats

1.9 Stock Indices by Country

Region	Indices
USA	S&P 500, Dow Jones Industrial (US30), Nasdaq 100 (NAS100), Nasdaq Composite, Russell 2000/1000/3000, S&P 400 MidCap, S&P 600 SmallCap, NYSE Composite, NYSE FANG+, Wilshire 5000, Philadelphia Semiconductor (SOX), Dow Transports, Dow Utilities, VIX, MOVE (bond vol)
Canada / Latin America	S&P/TSX Composite, TSX 60, Bovespa (Brazil), IPC (Mexico), Merval (Argentina), IPSA (Chile), COLCAP (Colombia), S&P/BVL Peru
UK / Ireland	FTSE 100, FTSE 250, FTSE All-Share, AIM, ISEQ 20
Eurozone	Euro Stoxx 50, Stoxx 600, DAX 40 (Germany), MDAX, SDAX, CAC 40 (France), AEX (Netherlands), BEL 20, IBEX 35 (Spain), FTSE MIB (Italy), PSI (Portugal), ATX (Austria), ATHEX (Greece), OMX Helsinki
Other Europe	SMI (Switzerland), OMX Stockholm 30, OBX (Norway), OMX Copenhagen 25, WIG20 (Poland), PX (Czech), BUX (Hungary), BIST 100 (Turkey), MOEX Russia, RTS, PFTS (Ukraine)
East Asia	Nikkei 225, TOPIX, Hang Seng, Hang Seng Tech, Shanghai Composite, SSE 50, Shenzhen Component, CSI 300, ChiNext, STAR 50, KOSPI, KOSDAQ, TAIEX
South / Southeast Asia	Nifty 50, Sensex, Bank Nifty, Nifty Midcap, KSE-100 (Pakistan), KSE-30, KMI-30, DSEX (Bangladesh), CSE All Share (Sri Lanka), NEPSE, STI (Singapore), KLCI (Malaysia), SET (Thailand), JCI (Indonesia), PSEi (Philippines), VN-Index
Oceania	ASX 200, All Ordinaries, NZX 50
Middle East / Africa	Tadawul (TASI, Saudi), DFM General, ADX General, QE Index (Qatar), Kuwait Main, Bahrain All Share, MSM 30 (Oman), TA-35 (Israel), EGX 30 (Egypt), JSE Top 40 (South Africa), NGX All Share (Nigeria), NSE 20 (Kenya), MASI (Morocco)
Global	MSCI World, MSCI ACWI, MSCI EM, MSCI Frontier, FTSE All-World, S&P Global 1200

1.10 Stock Exchanges of the World (major)

Region	Exchanges
North America	NYSE, Nasdaq, NYSE American, Cboe, TSX/TSXV, Mexican BMV
Europe	London SE, Euronext (Paris, Amsterdam, Brussels, Lisbon, Dublin, Milan, Oslo), Deutsche Boerse (Xetra, Frankfurt), SIX Swiss, Nasdaq Nordic, BME (Spain), Warsaw, Vienna, Athens, Borsa Istanbul, Moscow Exchange
Asia-Pacific	Japan Exchange Group (Tokyo), HKEX, Shanghai SE, Shenzhen SE, Beijing SE, Korea Exchange, Taiwan SE, NSE India, BSE India, Singapore Exchange, Bursa Malaysia, IDX Indonesia, SET Thailand, PSE Philippines, HOSE/HNX Vietnam, ASX, NZX, Pakistan Stock Exchange (PSX), Dhaka SE, Colombo SE
Middle East / Africa	Saudi Exchange (Tadawul), DFM, ADX, Qatar SE, Boursa Kuwait, Bahrain Bourse, Muscat, Tel Aviv SE, Egyptian Exchange, JSE (Johannesburg), Nigerian Exchange, Nairobi SE, Casablanca
Latin America	B3 (Brazil), Buenos Aires, Santiago, Bogota, Lima

1.11 Stocks by Sector (Examples)

Sector	Examples
Technology	Apple, Microsoft, Alphabet, Amazon, Meta, Nvidia, Tesla, Broadcom, TSMC, Samsung, ASML, Oracle, SAP, Adobe, Salesforce, AMD, Intel, Qualcomm, Cisco, IBM, Palantir, Tencent, Alibaba
Financials	JPMorgan, Bank of America, Goldman Sachs, Morgan Stanley, Visa, Mastercard, PayPal, HSBC, UBS, HBL, UBL, MCB, Meezan Bank, ICBC, Berkshire Hathaway
Energy	ExxonMobil, Chevron, Shell, BP, TotalEnergies, Saudi Aramco, OGDC, PPL, PSO, Schlumberger
Healthcare	Eli Lilly, Novo Nordisk, UnitedHealth, J&J, Pfizer, Merck, AbbVie, Roche, Novartis, AstraZeneca
Consumer	Walmart, Costco, Coca-Cola, PepsiCo, P&G, Nestle, Unilever, McDonald's, Nike, LVMH, Toyota, Ford, BYD, Engro, Lucky Cement, Fauji Fertilizer
Industrials / Defense	Boeing, Airbus, Caterpillar, GE, Lockheed Martin, RTX, Siemens, Honeywell

Sector	Examples
Telecom / Media	Verizon, AT&T, T-Mobile, Disney, Netflix, Comcast, PTCL, Jazz
Real Estate / Utilities	Prologis, American Tower, NextEra, Duke Energy, K-Electric, Hubco

Stock types: Common, Preferred, ADR/GDR, Dual-class, Penny stocks, Blue chip, Growth, Value, Dividend, IPO, SPAC, REIT shares, Tracking stocks, Rights issues, Bonus shares.

1.12 Cryptocurrencies and Digital Assets

Category	Examples
Store of value / Large cap	BTC, ETH, BNB, XRP, SOL, ADA, TRX, TON, AVAX, DOT, LTC, BCH, XLM, ETC, XMR, ZEC
Layer 1 / Smart contract	Ethereum, Solana, Cardano, Avalanche, Polkadot, Cosmos (ATOM), NEAR, Aptos, Sui, Algorand, Tezos, Hedera, Fantom/Sonic, Kaspa, Injective
Layer 2	Arbitrum, Optimism, Base, Polygon, zkSync, Starknet, Mantle, Linea, Scroll
DeFi	Uniswap, Aave, Maker/Sky, Curve, Lido, Compound, Synthetix, dYdX, GMX, PancakeSwap, Jupiter, Raydium
Stablecoins	USDT, USDC, DAI, USDe, FDUSD, PYUSD, TUSD, EURC, plus gold backed (PAXG, XAUT)
Meme coins	DOGE, SHIB, PEPE, FLOKI, BONK, WIF
AI / Compute	TAO, RENDER, FET, GRT, AKT, IO
Oracles / Infra	Chainlink, Pyth, Band, The Graph, Filecoin, Arweave
Gaming / Metaverse	AXS, SAND, MANA, GALA, IMX
Exchange tokens	BNB, OKB, CRO, KCS, LEO
NFTs	Art, collectibles, gaming items, domain names (ENS)
Tokenized / RWA	Tokenized treasuries, stocks, gold, real estate
Crypto Derivatives	Perpetual swaps, futures, options (Deribit), leveraged tokens, funding rate arbitrage
Crypto ETFs / Trusts	Spot Bitcoin and Ether ETFs, GBTC, Grayscale trusts
Indices	CMC 100, CoinDesk 20, Bitwise 10, Crypto Fear and Greed

1.13 Fixed Income (Bonds and Rates)

Type	Instruments
Government bonds	US Treasuries (T-Bills, 2Y, 5Y, 7Y, 10Y, 20Y, 30Y, TIPS, STRIPS), UK Gilts, German Bunds/Bobls/Schatz, French OATs, Italian BTPs, Spanish Bonos, JGBs, Canadian GoC, Australian ACGBs, Indian G-Secs, PIBs and T-Bills (Pakistan), Sukuk
Corporate	Investment grade, High yield (junk), Convertible, Floating rate notes, Perpetual, Green bonds, Sukuk (corporate)
Municipal / Agency	US Munis, GSE bonds (Fannie, Freddie), Supranationals (World Bank, ADB, EIB)
Securitized	MBS, ABS, CMO, CLO, CDO, Covered bonds
Rates and benchmarks	Fed Funds, SOFR, ESTR, SONIA, TONA, EURIBOR, KIBOR, KORIBOR, repo rates, prime rate
Rate derivatives	Interest rate swaps, FRAs, Eurodollar/SOFR futures, Bond futures (ZN, ZB, ZF, ZT, Bund, Gilt, JGB), Swaptions, Caps and Floors
Credit derivatives	CDS, CDX, iTraxx, Total return swaps
Inflation	TIPS, Breakeven rates, inflation swaps, I-bonds

1.14 Derivatives and Structured Products

Type	Description
Futures	Standardized, exchange traded (commodities, indices, FX, rates, crypto, volatility)
Options (Calls/Puts)	American, European, Bermudan styles on stocks, indices, ETFs, futures, FX
Exotic options	Barrier (knock-in/out), Digital/Binary, Asian, Lookback, Chooser, Compound, Rainbow, Basket, Quanto, Cliquet, Range accrual
Options strategies	Covered call, protective put, straddle, strangle, iron condor, iron butterfly, butterfly, calendar, diagonal, ratio spread, collar, bull/bear spreads, jade lizard
CFDs and Spread bets	Leveraged OTC contracts via brokers
Swaps	Interest rate, currency, commodity, equity, credit default, total return, variance, volatility
Forwards	OTC customized contracts

Type	Description
Warrants / Rights	Long dated equity options issued by companies
Structured notes	Capital protected notes, reverse convertibles, autocallables, participation notes
Binary / Fixed odds	One touch, high/low (regulated in some regions only)
Volatility products	VIX futures/options, VXX, UVXY, SVXY, variance swaps, VSTOXX, V-Nikkei, India VIX
Leveraged / Inverse ETFs	TQQQ, SQQQ, SPXL, SPXS, UPRO, SOXL
Spreads	Calendar, inter-commodity (crack, crush, spark), TED, NOB, butterfly

1.15 Funds and Pooled Instruments

Type	Examples
Index ETFs	SPY, VOO, IVV, QQQ, DIA, IWM, VTI, VT, EFA, EEM, VWO
Sector ETFs	XLK, XLF, XLE, XLV, XLY, XLP, XLI, XLU, XLB, XLRE, SMH, ARKK
Commodity ETFs	GLD, IAU, SLV, USO, UNG, DBC, PDBC, CPER, URA
Bond ETFs	TLT, IEF, SHY, AGG, BND, LQD, HYG, JNK, TIP, EMB
Crypto ETFs	IBIT, FBTC, ETHA, BITO
Mutual funds	Equity, bond, balanced, index, money market, target date
Closed-end funds, REITs	Property trusts (equity, mortgage, hybrid)
Hedge funds / Private equity / VC	Alternative pooled investments
Islamic funds	Shariah compliant equity and sukuk funds
Pension / Sovereign funds	Long horizon institutional vehicles

1.16 Alternative and Unusual Instruments

Instrument	Description
Real estate	Residential, commercial, land, REITs, crowdfunding, tokenized property
Art, wine, whisky, watches, classic cars, collectibles	Passion assets, platforms like Masterworks, Rally
Weather derivatives	HDD/CDD contracts (CME)
Freight / Shipping	Baltic Dry Index, FFA (Forward Freight Agreements), container rates (SCFI, Drewry)
Catastrophe bonds	Insurance linked securities
Carbon credits	Voluntary and compliance
Prediction markets	Event contracts (Kalshi, Polymarket)
Peer-to-peer lending, Invoice financing	Alternative credit
Royalties	Music and patent royalties
Commodities physical	Gold bars/coins, silver, diamonds
Domain names, Websites, SaaS businesses	Digital assets
Gaming / In-game items, Loyalty points	Niche digital
Life settlements	Insurance policy trading

PART 2: TECHNICAL INDICATORS (200+)

Format: Indicator, What it does, Typical default setting / signal.

2.1 Moving Averages and Overlays

Indicator	Purpose	Default / Signal
SMA	Simple average of closes	50, 100, 200. Price above = bullish
EMA	Recent price weighted	9, 12, 20, 21, 26, 50, 200
WMA	Linearly weighted	Fast response
SMMA / RMA / Wilder MA	Smoothed MA (used inside RSI, ATR)	14
HMA (Hull)	Low lag, smooth	9, 21
DEMA / TEMA	Double / triple EMA, less lag	20
KAMA (Kaufman Adaptive)	Adapts to noise	10, 2, 30
VIDYA	Variable index dynamic average	Volatility adaptive
FRAMA	Fractal adaptive MA	16
ALMA (Arnaud Legoux)	Gaussian weighted	9, 0.85, 6
T3 (Tillson)	Very smooth	5, 0.7
ZLEMA	Zero lag EMA	20
LSMA / Linear Regression Line	Least squares fit	25
VWMA	Volume weighted MA	20
McGinley Dynamic	Self adjusting MA	14
Jurik MA (JMA)	Smooth and fast	Proprietary
Displaced MA	Shifted forward/back	Shift 3-5
MA Ribbon / Guppy MMA	Many EMAs (3,5,8,10,12,15 and 30,35,40,45,50,60)	Trend compression/expansion
Golden Cross / Death Cross	50 SMA crossing 200 SMA	Long term trend signal
Bollinger Bands	SMA +/- 2 std dev	20, 2
Keltner Channel	EMA +/- 2 ATR	20, 10, 2
Donchian Channel	Highest high / lowest low	20 (Turtle breakout 20/55)
Envelopes	SMA +/- percent	20, 2.5%
Price Channels, Starc Bands	Range based channels	15, 6
Parabolic SAR	Trailing dots	0.02, 0.2
Supertrend	ATR trailing trend line	10, 3
Ichimoku Cloud	Tenkan, Kijun, Senkou A/B, Chikou	9, 26, 52
Chandelier Exit	ATR trailing stop from highs	22, 3
Kase Dev Stop, Volatility Stop	ATR stops	
Pivot Points (Classic, Fibonacci, Camarilla, Woodie, DeMark)	Daily/weekly levels	PP, R1-R3, S1-S3
VWAP, Anchored VWAP, VWAP Bands	Volume weighted average price	Session or anchored
Zig Zag	Filters swings	5% or 12/5/3
Heikin Ashi, Renko, Kagi, Point and Figure, Three Line Break, Range Bars	Alternate chart types	Noise reduction
Fractals (Williams)	Local highs/lows	5 bar
Linear Regression Channel	Trend +/- deviation	100, 2
Andrews Pitchfork	Median line and parallels	3 pivots

2.2 Trend Strength and Direction

Indicator	Purpose	Default / Signal
MACD	EMA12 - EMA26, signal 9, histogram	12, 26, 9
ADX	Trend strength (not direction)	14. Above 25 = strong trend

Indicator	Purpose	Default / Signal
DMI (+DI / -DI)	Direction	14
ADXR	Smoothed ADX	14
Aroon Up/Down/Oscillator	Time since high/low	25
Vortex Indicator (VI+/VI-)	Trend start	14
TRIX	Triple EMA rate of change	15
Mass Index	Range expansion reversals	25, 9. Reversal bulge 27/26.5
Alligator (Jaw, Teeth, Lips)	Trend vs sleeping	13/8, 8/5, 5/3
Gator Oscillator	Alligator convergence	-
Schaff Trend Cycle	Fast cycle trend	23, 50, 10
Choppiness Index	Trend vs range	14. Above 61.8 = choppy
Trend Intensity Index, Efficiency Ratio	Trend quality	30, 10
Hurst Exponent	Persistence / mean reversion	Above 0.5 trending
Qstick	Candle body average	8
Elder Impulse System	EMA + MACD hist coloring	13, 12-26-9
Know Sure Thing (KST)	Weighted ROC	10-15-20-30
Coppock Curve	Long term bottoms	14, 11, 10
Price Oscillator (APO/PPO)	MA difference	12, 26
DPO	Detrended price oscillator	20
Rainbow Oscillator, Trend Detection Index	Trend variants	
SuperTrend with ADX filter, Half Trend, UT Bot, Range Filter, Nadaraya-Watson Envelope	Popular custom trend tools	TradingView community

2.3 Momentum Oscillators

Indicator	Purpose	Default / Signal
RSI	Strength 0-100	14. 70/30 (80/20 in trends)
Stochastic (Fast/Slow/Full)	Close in range	14,3,3. 80/20
Stochastic RSI	RSI stochastic	14,14,3,3
CCI	Deviation from mean	20. +/-100, +/-200
Williams %R	Inverted stochastic	14. -20/-80
ROC / Momentum	Price change	12 / 10
Awesome Oscillator (AO)	5 vs 34 median SMA	5, 34
Accelerator Oscillator (AC)	AO acceleration	-
Ultimate Oscillator	3 timeframes	7, 14, 28
TSI	Double smoothed momentum	25, 13
CMO (Chande)	Momentum -100 to 100	14
Fisher Transform	Gaussian transform	9-10
Inverse Fisher Transform	Bounded RSI	-
Connors RSI	RSI + streak + percent rank	3, 2, 100
RVI (Relative Vigor)	Close vs open vs range	10
Elder Ray (Bull/Bear Power)	High/low vs EMA13	13
DeMarker (DeM)	Demand strength	14. 0.7/0.3
Stochastic Momentum Index (SMI)	Refined stochastic	10, 3, 3
Balance of Power (BoP)	Buyer vs seller power	14
Relative Momentum Index (RMI)	RSI with momentum lag	14, 5
Laguerre RSI	Smooth fast RSI	gamma 0.5
Klinger, PMO, Detrended Stoch	Other oscillators	
Ehlers: Cyber Cycle, Sine Wave, Stochastic CG, MESA Adaptive MA (MAMA/FAMA)	Cycle/DSP indicators	John Ehlers

Indicator	Purpose	Default / Signal
Squeeze Momentum (LazyBear)	BB inside KC plus momentum	20, 2, 1.5
Waddah Attar Explosion	Trend plus volatility	-
QQE, STC, WaveTrend, Vortex, Divergence indicators	Popular custom	MT4/TV

2.4 Volatility

Indicator	Purpose	Default / Signal
ATR	Average true range	14. Stop = 1.5-3x ATR
Bollinger Bandwidth, %B	Squeeze and position	20, 2
Standard Deviation	Dispersion	20
Historical / Realized Volatility	Annualized stdev	20, 30 days
Implied Volatility (IV), IV Rank, IV Percentile	Options priced volatility	-
VIX, VXN, OVX, GVZ, MOVE, VSTOXX	Fear gauges (equity, nasdaq, oil, gold, bonds, Europe)	VIX above 30 = fear
Chaikin Volatility	Range ROC	10, 10
Normalized ATR (NATR)	ATR percent	14
Ulcer Index	Downside volatility	14
Volatility Ratio, Relative Volatility Index (RVI)	Volatility strength	10, 14
True Range, Average Daily Range (ADR), Average Weekly Range	Range stats	14, 20
Parkinson, Garman-Klass, Yang-Zhang	Range-based volatility estimators	Quant
GARCH models	Volatility forecasting	Quant

2.5 Volume and Money Flow

Indicator	Purpose	Default / Signal
Volume, Volume MA	Activity	20
OBV	Cumulative signed volume	Divergence
Accumulation/Distribution	Close location x volume	-
Chaikin Money Flow (CMF)	Buy/sell pressure	20
Chaikin Oscillator	MACD of A/D	3, 10
Money Flow Index (MFI)	Volume RSI	14. 80/20
VWAP	Fair value	Session
Volume Profile (VPVR, VPSV, Session Volume Profile)	Volume by price	POC, VAH, VAL
Force Index	Price change x volume	13
Ease of Movement	Move per volume	14
Klinger Volume Oscillator	Long term flow	34, 55, 13
Price Volume Trend (PVT)	Cumulative PV	-
Negative/Positive Volume Index	Smart vs dumb money	255 EMA
Volume Oscillator	Fast vs slow volume MA	5, 10
Market Facilitation Index (BW MFI)	Range per volume (4 color bars)	Green/Fade/Fake/Squat
Delta, Cumulative Delta, Footprint, CVD	Order flow	Bid vs ask volume
Depth of Market (DOM), Level 2, Time and Sales, Heatmap (Bookmap)	Order book tools	Futures/crypto
Tick Volume, Real Volume	Forex volume proxies	MT4 uses tick
Open Interest	Outstanding contracts	Futures/options
Volume Weighted RSI, Twiggs Money Flow, Intraday Intensity	Variants	

Indicator	Purpose	Default / Signal
Anchored VWAP, TPO Value Area	Auction levels	

2.6 Support, Resistance, Cycles and Geometry

Tool	Purpose	Levels / Notes
Horizontal S/R, Trendlines, Channels	Basic structure	Draw on swings
Fibonacci Retracement	Pullback levels	23.6, 38.2, 50, 61.8, 78.6
Fibonacci Extension / Projection	Targets	61.8, 100, 127.2, 161.8, 261.8
Fibonacci Fan, Arcs, Channel, Time Zones, Spiral	Fib geometry	
Gann Fan, Gann Box, Square of 9, Square of 144, Hexagon, Angles (1x1, 2x1, etc.)	Gann geometry	W.D. Gann
Murrey Math Lines	8/8 to 0/8 grid	-
Pivot Points (see overlays)	Levels	
Round numbers, Psychological levels	00 and 50 levels	
Supply and Demand Zones, Order Blocks	Institutional zones	
Previous Day/Week/Month High, Low, Close, Open	Reference levels	
Opening Range (ORB)	First 15-60 min range	Breakout
Initial Balance (IB)	First hour range	Market profile
Time cycles, Lunar cycles, Seasonality, Presidential cycle	Time analysis	
Elliott Wave tools, Fibonacci time	Wave counts	
Regression channels, Standard error bands	Statistical	
Speed Resistance Lines, Quadrant Lines, Cycle Lines, Tirone Levels	Other drawing tools	

2.7 Breadth, Sentiment and Positioning

Indicator	Purpose
Advance/Decline Line, A/D Ratio	Market breadth
McClellan Oscillator and Summation Index	Breadth momentum
TRIN / Arms Index, TICK Index	Intraday breadth
% of stocks above 50/200 DMA	Participation
New Highs / New Lows, High-Low Index	Breadth
Up/Down Volume	Volume breadth
Put/Call Ratio, Skew Index (CBOE SKEW)	Options sentiment
AAII Sentiment, Investors Intelligence, CNN Fear and Greed, Crypto Fear and Greed	Survey / composite sentiment
COT Report (Commitment of Traders), TFF, Disaggregated	Futures positioning (commercials vs speculators)
Retail Positioning (OANDA/IG/MyFXBook/Dukascopy SWFX)	Retail long/short ratios (contrarian)
Margin Debt, Fund Manager Survey (BofA), Short Interest, Days to Cover	Equity positioning
Funding Rate, Open Interest, Long/Short Ratio, Liquidations, Basis	Crypto derivatives sentiment
Google Trends, Social volume, News sentiment (NLP)	Alternative data
DXY, Currency Strength Meter, Correlation Matrix	Intermarket

2.8 Crypto On-chain Indicators

Indicator	Purpose
MVRV, MVRV Z-Score, NUPL, SOPR, Realized Price	Profit/loss of holders
Hash Rate, Difficulty, Hash Ribbons, Miner Outflow	Miner health
Exchange Netflow, Exchange Reserve	Sell/buy pressure

Indicator	Purpose
Active Addresses, New Addresses, Transaction Count, Gas fees	Network usage
Whale Transactions, Holder distribution	Large players
Stock-to-Flow, Rainbow Chart, Pi Cycle Top, Puell Multiple, Mayer Multiple	Cycle models
Bitcoin Dominance (BTC.D), ETH/BTC, Altcoin Season Index, Total Market Cap	Market structure
Stablecoin Supply Ratio (SSR), Stablecoin Market Cap	Dry powder
TVL (Total Value Locked), DEX volume, Protocol revenue	DeFi health
Coin Days Destroyed, Dormancy, Liveliness, Reserve Risk	HODLer behavior

PART 3: PATTERNS

3.1 Japanese Candlestick Patterns

Type	Patterns
Single candle	Doji (standard, long-legged, dragonfly, gravestone), Hammer, Inverted Hammer, Hanging Man, Shooting Star, Spinning Top, Marubozu (white/black), High Wave, Belt Hold
Bullish reversal (2-3 candle)	Bullish Engulfing, Piercing Line, Bullish Harami, Harami Cross, Tweezer Bottom, Morning Star, Morning Doji Star, Three White Soldiers, Three Inside Up, Three Outside Up, Bullish Kicker, Homing Pigeon, Matching Low, Unique Three River Bottom, Abandoned Baby (bullish), Ladder Bottom
Bearish reversal	Bearish Engulfing, Dark Cloud Cover, Bearish Harami, Tweezer Top, Evening Star, Evening Doji Star, Three Black Crows, Three Inside Down, Three Outside Down, Bearish Kicker, Abandoned Baby (bearish), Advance Block, Deliberation, Upside Gap Two Crows, Identical Three Crows
Continuation	Rising Three Methods, Falling Three Methods, Mat Hold, Separating Lines, Side-by-Side White Lines, Upside/Downside Tasuki Gap, Upside/Downside Gap Three Methods, In-Neck, On-Neck, Thrusting
Gap patterns	Breakaway Gap, Runaway/Measuring Gap, Exhaustion Gap, Common Gap, Island Reversal

3.2 Classical Chart Patterns

Type	Patterns
Reversal	Head and Shoulders, Inverse H&S, Double Top/Bottom, Triple Top/Bottom, Rounding Top/Bottom (Saucer), V-Top/V-Bottom, Diamond Top/Bottom, Broadening Top, Island Reversal, Rising/Falling Wedge, Adam and Eve
Continuation	Bull/Bear Flag, Pennant, Ascending Triangle, Descending Triangle, Symmetrical Triangle, Rectangle (Range), Cup and Handle, Ascending/Descending Channel, Measured Move, Three Drives
Special	Broadening Formation (Megaphone), Triangle Expanding, Wolfe Waves, Head and Shoulders Complex, Bump and Run

3.3 Harmonic Patterns

Pattern	Key Fibonacci ratios
Gartley	B 0.618 of XA, D 0.786 of XA
Bat	B 0.382-0.5, D 0.886 of XA
Alternate Bat	B 0.382, D 1.13 of XA
Butterfly	B 0.786, D 1.27-1.618 of XA
Crab	B 0.382-0.618, D 1.618 of XA
Deep Crab	B 0.886, D 1.618 of XA
Shark	0.886-1.13 of OX, 5-0 relation
Cypher	B 0.382-0.618 of XA, C 1.272-1.414 of XA, D 0.786 of XC
ABCD	CD = AB (equal legs) or 1.272/1.618 of BC
Three Drives	Drives at 1.272 or 1.618 extensions
5-0, Nen Star, Anti patterns, Black Swan, White Swan	Advanced harmonic variants

3.4 Price Action / Smart Money Concepts

Concept	Meaning
Market Structure: HH, HL, LH, LL	Trend definition
BOS (Break of Structure)	Trend continuation break
CHoCH (Change of Character) / MSS	First sign of reversal
Order Block (OB), Breaker Block, Mitigation Block	Last opposite candle before impulsive move
Fair Value Gap (FVG) / Imbalance / Inefficiency	3-candle gap price tends to revisit
Liquidity (Buy-side / Sell-side), Equal Highs/Lows, Stop Hunt, Liquidity Sweep	Resting stop orders
Premium / Discount, Equilibrium (50%)	Range pricing
Optimal Trade Entry (OTE)	62-79% retracement
Inducement, Turtle Soup, Judas Swing	Fake moves

Concept	Meaning
Killzones (London, NY AM/PM, Asia)	High probability time windows
Power of 3 (Accumulation, Manipulation, Distribution)	Daily range model
Supply/Demand, Rally-Base-Rally, Drop-Base-Drop	Zone theory
Pin Bar, Inside Bar, Outside Bar, Fakey, 2B pattern, Spring/Upthrust	Price action setups (Nial Fuller, Al Brooks)

PART 4: TRADING METHODOLOGIES AND THEORIES

Method	Key idea
Dow Theory	Primary, secondary, minor trends. Averages must confirm. Volume confirms trend.
Elliott Wave	5 impulse waves + 3 corrective (ABC). Fibonacci relations. Patterns: impulse, diagonal (leading/ending), zigzag, flat, triangle, WXY combinations.
Wyckoff Method	Accumulation (PS, SC, AR, ST, Spring, SOS, LPS) and Distribution (PSY, BC, AR, UT, UTAD, SOW, LPSY). Composite man, cause and effect.
Gann Theory	Time/price squares, angles, cycles, Square of 9
Ichimoku Kinko Hyo	Cloud based complete system
Market Profile (TPO) and Auction Market Theory	Value area, POC, balanced vs imbalanced days
Volume Spread Analysis (VSA)	Tom Williams. Effort vs result
Price Action Trading	Naked chart, S/R, patterns
Smart Money Concepts / ICT	Inner Circle Trader methods (see 3.4)
Supply and Demand (Sam Seiden)	Institutional order zones
Turtle Trading	Donchian 20/55 breakouts, ATR sizing
Trend Following	Moving average systems, CTAs
Mean Reversion	RSI/BB extremes, pairs trading, stat arb
Breakout / Momentum Trading	Range expansion
Scalping, Day trading, Swing, Position trading	Timeframe styles
Grid and Martingale systems	Averaging (high risk)
Carry Trade	Borrow low-yield, hold high-yield currency
Arbitrage	Triangular, statistical, cross-exchange, funding
Algorithmic / HFT / Market Making	Automated systems
Machine Learning / AI Models	Regression, trees, LSTM, transformers, reinforcement learning
Quantitative Factor Models	Value, momentum, quality, size, low vol (Fama-French)
Fundamental Analysis	DCF, ratios, macro
Sentiment and Contrarian	Fade the crowd
Intermarket Analysis (John Murphy)	Bonds, stocks, commodities, USD relationships
Seasonality, Calendar Effects	Sell in May, Santa Rally, January effect, turn of month
Event / News Trading	NFP, CPI, FOMC, earnings
Renko, Point and Figure, Kagi systems	Time-independent charting
Fibonacci, Harmonic Trading (Scott Carney)	Ratio based
Heikin Ashi trend riding, Pivot Boss, CPR (Central Pivot Range)	Popular retail systems

PART 5: FUNDAMENTAL AND ECONOMIC INDICATORS

Category	Indicators
Growth	GDP (QoQ, YoY), GDP Nowcast (Atlanta Fed), PMI (Manufacturing, Services, Composite: S&P Global, ISM, Caixin, Au Jibun), Industrial Production, Capacity Utilization, Factory Orders, Durable Goods, Retail Sales, Business Inventories, Leading Economic Index
Inflation	CPI, Core CPI, PPI, PCE, Core PCE, Import/Export Prices, Employment Cost Index, Inflation Expectations (UMich, Breakevens), Trimmed Mean, Wage Growth
Labor	Non-Farm Payrolls, Unemployment Rate, U-6, Participation Rate, ADP, Jobless Claims (Initial/Continuing), JOLTS, Average Hourly Earnings, Employment Change (AU, CA, UK), Claimant Count
Central Banks	Fed (FOMC, Dot Plot, Minutes, Beige Book, Jackson Hole), ECB, BoE, BoJ, SNB, RBA, RBNZ, BoC, SBP, RBI, PBoC, CBRT, SARB, Banxico, BCB. Rate decisions, QE/QT, forward guidance, yield curve control

Category	Indicators
Trade and External	Trade Balance, Current Account, Capital Flows (TIC), FX Reserves, Remittances (key for Pakistan), Exports/Imports, Terms of Trade, FDI
Fiscal	Budget deficit, Debt/GDP, Treasury auctions (bid-to-cover), Sovereign ratings (Moody's, S&P, Fitch), CDS spreads
Consumer and Business	Consumer Confidence (CB, UMich), ZEW, Ifo, Sentix, Tankan, NFIB, Empire State, Philly Fed, Richmond Fed, Chicago PMI, Beige Book
Housing	Housing Starts, Building Permits, Existing/New Home Sales, Pending Sales, Case-Shiller, NAHB, Mortgage Rates, MBA Applications
Money and Credit	M0, M1, M2, M3, Bank Lending, Credit Spreads, Yield Curve (2s10s, 3m10y), TED Spread, Libor-OIS
Commodity Reports	EIA Crude Inventories, API, Baker Hughes Rig Count, OPEC MOMR, WASDE, Crop Progress, Natural Gas Storage, LME Stocks, China customs data
Stock Fundamentals	EPS, Revenue, P/E, Forward P/E, PEG, P/B, P/S, EV/EBITDA, ROE, ROA, ROIC, Debt/Equity, Current Ratio, Interest Coverage, Dividend Yield, Payout Ratio, FCF Yield, Gross/Operating/Net Margin, Beta, Short Interest, Insider Buying, Altman Z-Score, Piotroski F-Score
Crypto Fundamentals	Market cap, FDV, Circulating vs Total supply, Tokenomics, Unlock schedules, Developer activity, TVL, Revenue, Active users

PART 6: INTERMARKET RELATIONSHIPS (Rules of Thumb)

Relationship	Typical behavior (can change)
USD vs Gold	Usually inverse
USD vs EUR/USD	Strongly inverse (EUR is 57% of DXY)
Oil vs CAD	Higher oil tends to support CAD (USD/CAD down)
Oil vs inflation	Oil up raises inflation expectations
Gold vs real yields (TIPS)	Inverse
US 10Y yield vs USD/JPY	Positively correlated
AUD vs China data, Copper, Iron ore	Positive
Risk-on	Stocks, AUD, NZD, crypto up; JPY, CHF, gold, USD (sometimes) down
Risk-off	Reverse of risk-on; VIX up
Bonds vs stocks	Often inverse, breaks in inflation shocks
BTC vs Nasdaq	Often positively correlated
Silver vs Gold (Gold/Silver ratio)	Ratio rises in fear
Copper/Gold ratio	Growth vs safety
Yield curve inversion	Recession signal historically
EUR/USD vs GBP/USD	High positive correlation
USD/CHF vs EUR/USD	High negative correlation

PART 7: RISK, PERFORMANCE METRICS, ORDERS, SESSIONS

7.1 Order Types

Order	Meaning
Market	Immediate at best price
Limit	At price or better
Stop / Stop-Loss	Triggers market order at level
Stop-Limit	Triggers limit order
Buy Stop / Sell Stop / Buy Limit / Sell Limit	Pending orders in MT4
Take Profit (TP)	Exit at target
Trailing Stop	Moves with price
OCO (One Cancels Other)	Linked orders
Bracket	Entry + TP + SL
IOC / FOK / GTC / GTD / Day	Time in force

Order	Meaning
Iceberg, TWAP, VWAP orders	Algorithmic execution
Break-even stop	SL moved to entry
Partial close, Scale in/out	Position management

7.2 Risk and Money Management Metrics

Metric	Meaning
Risk per trade	Typically 0.5-2% of equity
Position size	Risk amount / (stop distance x pip value)
Risk:Reward (R:R)	Target distance / stop distance
Expectancy	(Win% x Avg Win) - (Loss% x Avg Loss)
Kelly Criterion, Fractional Kelly	Optimal bet fraction
Fixed Fractional, Fixed Ratio, Anti-Martingale	Sizing models
Leverage, Margin, Margin Level, Free Margin, Margin Call, Stop Out	Broker mechanics
Pip, Pipette, Point, Tick, Lot (Standard 100k, Mini, Micro, Nano)	Units
Spread, Commission, Swap/Rollover, Slippage	Costs
Correlation risk, Exposure, VaR, CVaR/Expected Shortfall	Portfolio risk

7.3 Performance Metrics

Metric	Meaning
Win Rate	Percent winning trades
Profit Factor	Gross profit / gross loss (above 1.5 good)
Max Drawdown, Average Drawdown, Recovery Factor	Equity decline measures
Sharpe Ratio	Return per total volatility
Sortino Ratio	Return per downside volatility
Calmar Ratio	CAGR / Max drawdown
Omega, Information Ratio, Treynor	Other ratios
CAGR, Monthly returns, Equity curve slope	Growth
MAE / MFE	Max adverse / favorable excursion
Walk-forward, Monte Carlo, Out-of-sample testing	Backtest validation
Overfitting, Curve fitting, Look-ahead bias, Survivorship bias	Backtest pitfalls

7.4 Trading Sessions (approx. GMT)

Session	Hours	Notes
Sydney	21:00 - 06:00	Quiet, AUD/NZD
Tokyo / Asia	00:00 - 09:00	JPY, AUD, range moves
London	07:00 - 16:00	Highest volume, trends start
New York	12:00 - 21:00	USD news, overlaps London 12-16
Karachi (PKT = GMT+5)	Add 5 hours	London open about 12:00 PKT, NY open about 18:00 PKT

PART 8: MT4 / MT5 INDICATORS

8.1 Built-in (Standard)

Group	Indicators
Trend	Average Directional Movement Index, Bollinger Bands, Envelopes, Ichimoku Kinko Hyo, Moving Average, Parabolic SAR, Standard Deviation (MT5 adds DEMA, TEMA, FRAMA, VIDYA, AMA)
Oscillators	ATR, Bears Power, Bulls Power, CCI, DeMarker, Force Index, MACD, Momentum, Moving Average of Oscillator (OsMA), RSI, Relative Vigor Index, Stochastic, Williams %R, TriX (MT5)
Volumes	Accumulation/Distribution, Money Flow Index, On Balance Volume, Volumes
Bill Williams	Accelerator, Alligator, Awesome, Fractals, Gator, Market Facilitation Index
Examples (MT5)	ADXW, Chaikin Oscillator, Bands, Volumes, Price Channel

8.2 Popular Custom MT4 Indicators (free community)

Indicator	Use
Super Trend, Half Trend, Trend Magic, Hull MA colored	Trend direction
Currency Strength Meter, Correlation Matrix	Pair selection
Supply and Demand Zones, Support Resistance auto, Pivot Level auto	Zones
ZigZag with Fibonacci auto, Harmonic Pattern Scanner	Patterns
Order Block / FVG indicators, Smart Money Concept dashboards	SMC
Session Boxes (Asia/London/NY), Market Clock, Spread/ATR panel	Context
Volume Profile, Order Flow (MT5 DOM), Market Profile	Advanced
News Calendar / Economic news indicators	Events
Trade Panels, Risk Calculator, Lot Size Calculator, One-click trading EAs	Execution helpers
Expert Advisors (EAs): grid, scalper, trend EA, news EA, copy trader, signal copier	Automation (test on demo only)

Disclaimer: Educational purpose only. Trading mein capital loss ka risk hai. Pehle demo account par practice karein, aur sirf utna risk lein jo afford kar saken.